

# Local affine-isometry chart on a smaller ball

MathlibAnnex.IsometryEquiv.exists\_affineExtension\_eqOn\_ball

## theorem

Constructs one ambient surjective affine isometry from the same radial map throughout.

## Statement

Let  $E, F$  be real normed spaces,  $S \subseteq E$ ,  $T \subseteq F$ , and  $f : S \rightarrow T$  a bijective isometry. Suppose  $c \in S$ ,  $R > 0$ ,  $B(c, R) \subseteq S$  and  $B(f(c), R) \subseteq T$ . There exists a surjective real affine isometry  $A : E \rightarrow F$  such that

$$A(x) = f(x) \quad (x \in S, \|x - c\| < R/8).$$

## Assumptions

Both ambient radius- $R$  balls must lie in the corresponding subsets. No finite dimensionality or completeness is assumed for either real normed space.

## Conclusion

One ambient affine isometry agrees with  $f$  on the smaller ball  $B(c, R/8)$ . The quantified assertion is existence; its agreement here is not asserted on all of  $S$ .

## Proof route

Extend the translated local map radially, prove that this same map is a linear isometry, obtain surjectivity from the target ball, and translate back.

### ▼ Proof steps

1. Set  $r = R/8$ ; this is the radius used throughout the construction. In Radial-map definition, define  $L_c(0) = 0$  and, for  $v \neq 0$ ,

$$L_c(v) = \frac{\|v\|}{r} \left[ f\left(c + \frac{r}{\|v\|}v\right) - f(c) \right].$$

The evaluation point is at distance  $r < R$  from  $c$ , hence belongs to  $S$ .

2. For  $a > 0$ , the evaluation point for  $av$  equals that for  $v$ , and the prefactor is multiplied by  $a$ . Thus Positive homogeneity of the radial map gives  $L_c(av) = aL_c(v)$ . Distance preservation gives Norm of the radial map in the concrete form

$$\|L_c(v)\| = \frac{\|v\|}{r} \left\| f\left(c + \frac{r}{\|v\|}v\right) - f(c) \right\| = \frac{\|v\|}{r} r = \|v\|.$$

The case  $v = 0$  follows from the definition.

3. If  $0 < \|v\| < r$ , first write the point  $c + v$  as the explicit convex combination

$$c + v = \left(1 - \frac{\|v\|}{r}\right)c + \frac{\|v\|}{r}\left(c + \frac{r}{\|v\|}v\right).$$

The coefficient  $\|v\|/r$  lies in  $(0, 1)$ , and both endpoints lie in  $B(c, R/4)$  because  $r = R/8$ . Apply Quarter-ball segment theorem to these two endpoints. It gives

$$f(c + v) - f(c) = \frac{\|v\|}{r} \left[ f\left(c + \frac{r}{\|v\|}v\right) - f(c) \right] = L_c(v).$$

The equality also holds for  $v = 0$ . This is Local agreement with the translated isometry.

4. For  $\|v\|, \|w\| < r$ , their midpoint also has norm less than  $r$ . Substitute the local agreement into Quarter-ball midpoint theorem for  $c + v, c + w$  to obtain

$$L_c((v + w)/2) = (L_c(v) + L_c(w))/2.$$

This verifies the local identity in Local midpoint identity for the radial map. For arbitrary  $v, w$ , choose  $\lambda = r/(\|v\| + \|w\| + 1) > 0$ . Both  $\lambda v, \lambda w$  lie in  $B(0, r)$ . Apply the local identity to these two vectors and use positive homogeneity for the same  $L_c$ :

$$\begin{aligned} \lambda L_c((v + w)/2) &= L_c((\lambda v + \lambda w)/2) \\ &= (L_c(\lambda v) + L_c(\lambda w))/2 \\ &= \lambda(L_c(v) + L_c(w))/2. \end{aligned}$$

Since  $\lambda > 0$ , division by  $\lambda$  gives the midpoint identity for the original  $v, w$ . This is the application of Rescaling local midpoint identities that proves the global midpoint identity.

5. Putting  $w = 0$  in the midpoint identity gives  $L_c(v/2) = L_c(v)/2$ . It follows that  $L_c(v + w) = L_c(v) + L_c(w)$ . Consequently

$$\|L_c(v) - L_c(w)\| = \|L_c(v - w)\| = \|v - w\|.$$

Thus  $L_c$  is continuous and additive, hence real-linear by rational approximation. The same midpoint map as a linear isometry equips this same function with the linear-isometry structure; it does not replace it by a different map.

6. Let  $\|y\| < r$ . The target ball inclusion puts  $f(c) + y$  in  $T$ . Take  $x = f^{-1}(f(c) + y)$ . Isometry gives  $\|x - c\| = \|y\| < r$ , so local agreement yields

$$L_c(x - c) = f(x) - f(c) = y.$$

Hence  $B(0, r) \subseteq \text{ran } L_c$ . This range is a real vector subspace. For arbitrary  $y$ , the scalar  $b = r/(2(\|y\| + 1)) > 0$  puts  $by$  in that ball; if  $L_c(v) = by$ , then  $L_c(v/b) = y$ . Thus  $L_c$  is surjective, without any dimension argument.

7. Define  $A(x) = L_c(x - c) + f(c)$ . Real linearity, norm preservation and surjectivity of the same  $L_c$  make  $A$  a surjective affine isometry. When  $\|x - c\| < r$ , substitute  $v = x - c$  into local agreement to obtain  $A(x) = f(x)$ .

## Main citations

- Exact theorem and proof
- Radial-map definition
- Positive homogeneity of the radial map
- Norm of the radial map
- Local agreement with the translated isometry
- Local midpoint identity for the radial map
- Rescaling local midpoint identities
- The same midpoint map as a linear isometry
- Quarter-ball segment theorem
- Quarter-ball midpoint theorem

## Lean source signature (exact)

```
theorem exists_affineExtension_eqOn_ball
  {s : Set E} {t : Set F} (f : s ≈i t) (c : s) {R : ℝ} (hR : 0 < R)
  (hsball : ball (c : E) R ⊆ s)
  (htball : ball ((f c : t) : F) R ⊆ t) :
  ∃ A : E ≈a i[ℝ] F, ∀ x : s,
    (x : E) ∈ ball (c : E) (R / 8) → A (x : E) = ((f x : t) : F)
```

| In the source                                           | Mathematical meaning                                                                                                                      |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| $(f : s \approx_i t) (c : s)$                           | The distance-preserving bijection $f : S \rightarrow T$ and center $c \in S$ , with $s \subseteq E, t \subseteq F$ in real normed spaces. |
| $\{R : \mathbb{R}\} (hR : 0 < R)$                       | The common radius $R > 0$ .                                                                                                               |
| $(hsball : \text{ball } (c : E) R \subseteq s)$         | The open ball $B(c, R)$ lies in the source set.                                                                                           |
| $(htball : \text{ball } ((f c : t) : F) R \subseteq t)$ | The open ball $B(f(c), R)$ lies in the target set; this is a separate hypothesis.                                                         |
| $\exists A : E \approx^{a i}[\mathbb{R}] F$             | There exists a surjective real affine isometry $A$ between the whole ambient spaces $E, F$ .                                              |
| $\forall x : s$                                         | After choosing one $A$ , the conclusion is required for every point $x \in S$ .                                                           |
| $(x : E) \in \text{ball } (c : E) (R / 8)$              | The additional condition on that point is $\ x - c\  < R/8$ .                                                                             |
| $A (x : E) = ((f x : t) : F)$                           | At each such point, the two ambient values agree: $A(x) = f(x)$ . The casts retain the same $x$ and $f(x)$ .                              |

Exact source and proof.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.IsometryEquiv.exists_affineExtension_eqOn_ball`

Accepted content SHA-256: `1dfa395e4a0993253ed4a48829331d9c687da7154627e150226d684d94ead2c5`

Accepted source guide SHA-256: `963c571770da523baa90c7fc770e157436055dcb00b1c1d5e569aed54b966b12`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`